

End-to-End Data Cleaning

Problem Statement:

A financial institution is analyzing data from credit applications to predict the credit risk of applicants. This data contains both numerical and categorical features such as age, job type/change, housing status, and account information.

The institution wants to clean and prepare the data for further analysis and machine learning modeling.

The original dataset contains 1000 entries with 20 categorical/symbolic attributes prepared by Prof. Hofmann.

In this dataset, each entry represents a person who takes a credit by a bank.

Each person is classified as good or bad credit risks according to the set of attributes.

It is almost impossible to understand the original dataset due to its complicated system of categories and symbols. Therefore, a simplified version is prepared where several columns are ignored and simplified.

The selected attributes are:

Feature Name	Type	Description
Age	Numeric	Age of the person (in years).
Sex	Categorical	Gender of the person (male , female).
Job	Ordinal	Job classification: 0 = Unskilled and non-resident 1 = Unskilled and resident 2 = Skilled 3 = Highly skilled
Housing	Categorical	Housing situation (own , rent , or free).
Saving accounts	Ordinal	Amount in savings account: little , moderate , quite rich , or rich .
Checking account	Numeric	Amount in checking account (in Deutsch Marks - DM).
Credit amount	Numeric	Amount of credit taken (in DM).
Duration	Numeric	Duration of the credit (in months).
Risk	Categorical	Risk score for credit application (good or bad)

Link to download the dataset: [credit_data.csv](#) or visit the github repo of the course https://github.com/benouinrachid/DQDC_2025

Objective:

Your task is to clean the provided dataset and provide and train a simple ML model for predicting Credit Risk.

- **Handle missing values** appropriately.
- **Apply suitable encoding** to categorical variables using:
 - **Ordinal encoding** for features with an inherent order.
 - **Label encoding** for nominal categorical features.
- **Check for outliers** in numerical features and decide on a strategy to handle them (e.g., capping, removal).
- **Check for class imbalance** in the target variable (Risk) and suggest ways to address it if needed (e.g., resampling techniques).
- **Apply feature scaling or normalization** where appropriate to prepare data for modeling.
- **Train a simple machine learning model** (e.g., Decision Tree, Logistic Regression, etc.) to predict the Risk.
 - **Justify the choice of the model** based on the nature of the data and problem.
- **Evaluate the model's performance** using appropriate metrics (e.g., accuracy, f1-score, precision, recall).
- **Document your steps and decisions** clearly as part of your submission (PDF file).

Deliverable:

Students must submit a PDF Report as follows:

- A well-organized report
- Explanation of each preprocessing step
- Plots showing:
 - Class distribution
 - Boxplots or histograms for outlier detection
- Any relevant correlation or feature distribution insights
- Description of the model and its justification of use

- Evaluation results and discussion
- Conclusion and recommendations
- **Optional:** .csv file of the cleaned and encoded dataset

Deadline:

- To submit before **13:00 - Friday, April 18, 2025**